Sorting

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Sorting

Arrangement of data in a particular order.

$$\begin{array}{cccccc} & 0 & 1 & 2 & 3 & 4 \\ A = [& 2 & -5 & 8 & 10 & -12] \end{array}$$

sorted wrt abs (A[i]) $|-5| = 5$

$$\begin{array}{cccccc} & 0 & 1 & 2 & 3 & 4 \\ A = [& 1 & 13 & 9 & 6 & 12] \end{array}$$

\# of factors → 1 2 3 4 6

Question (Elements Removal)

Given N elements, at every step remove an array element.

Cost to remove an element = Sum of array of elements present in an array

Find minimum cost to remove all elements.

NOTE : First add the cost of removal and then remove it.

arr - [2 1 4]

Remove	cost	Rem. Array
2	2+1+4 = 7	[1 4]
1	1+4 = 5	[4]
4	4	[ ]
	16 ✗	

4	2 + 1 + 4 = 7	[2 1]
2	2 + 1 = 3	[1]
1	1	[ ]
	11 ✓	

A = [4 6 1]

Remove	Cost	A
6	4 + 6 + 1 = 11	[4 1]
4	4 + 1 = 5	[1]
1	1	[ ]
	17	

$$A = \begin{array}{cccc} 0 & 1 & 2 & 3 \\ \hline [3 & 5 & 1 & -3] \end{array}$$

cost	Remove	A
3 + 5 + 1 − 3 = 6	5	[3 1 −3]
3 + 1 − 3 = 1	3	[1 −3]
1 − 3 = −2	1	[−3]
−3	−3	[ ]
2		

Observation → Remove elements in descending order to get min cost. ✓

A = [a b c]

cost	Remove
a + b + c	a
b + c	b
c	c

Descending Order

1a + 2b + 3c

To keep it min. $a \geq b \geq c$

```
// Sort A in desc. order
ans = 0
for i → 0 to (N-1) {
    ans += A[i] * (i+1)
}
return ans
```

TC = O(N log(N))

SC = O(1)

Question (Noble Integers) { Distinct data }

Given N array elements, calculate number of noble integers.

An element ele in arr [ ] is said to be noble if { count of smaller elements = ele itself }

0 1 2 3 4 5

arr - [1 , -5 , 3 , 5 , -10 , 4]

elements < A[i] → 2 1 3 5 0 4 Ans = 3

0 1 2 3

arr - [-3 , 0 , 2 , 5]

elements < A[i] → 0 1 2 3 Ans = 1

Bruteforce → ∀ i, count # elements < A[i] & compare it with A[i].

TC = O(N²) SC = O(1)

count # elements < A[i] → Rearrange s.t ∀A[i] smaller elements are on the left

⇒ Ascending order sorting

(# elements < A[i]) = i

```
// sort A[] in asc. order
ans = 0
for i → 0 to (N-1) {
|     if (A[i] == i)     ans++
}
```

$TC = \underline{O(N \log(N))}$

$SC = \underline{O(1)}$

$$\begin{array}{rccc} i \rightarrow & 0 & 1 & 2 \\ A = [& 1 & 2 & 3 \;] \end{array}$$

Ans = $\underline{0}$

Question (Noble Integers) : { Data can repeat }

	0	1	2	3	4
arr -	-10	1	1	3	100
# e < A[i] →	0	1	1	3	4

Ans = 3

	0	1	2	3	4	5	6	7	8
arr -	-10	1	1	2	4	4	4	8	10
# e < A[i] →	0	1	1	3	4	4	4	7	8

Ans = 5

	0	1	2	3	4	5	6	7	8	9	10	11	12	13
arr -	-3	0	2	2	5	5	5	5	8	8	10	10	10	14
# e < A[i] →	0	1	2	2	4	4	4	4	8	8	10	10	10	13

Ans = 7

```
// Sort A[] in ascending order
ans = 0            cnt = 0
for i → 0 to (N-1) {
    if ( i != 0 && A[i] == A[i-1])
        // no change in cnt
    else   cnt = i
    if ( A[i] == cnt )   ans++
}
return ans
```

TC = O(N log(N))

SC = O(1)

Selection Sort

idea : Select the minimum element and send that elements to correct position by swapping.

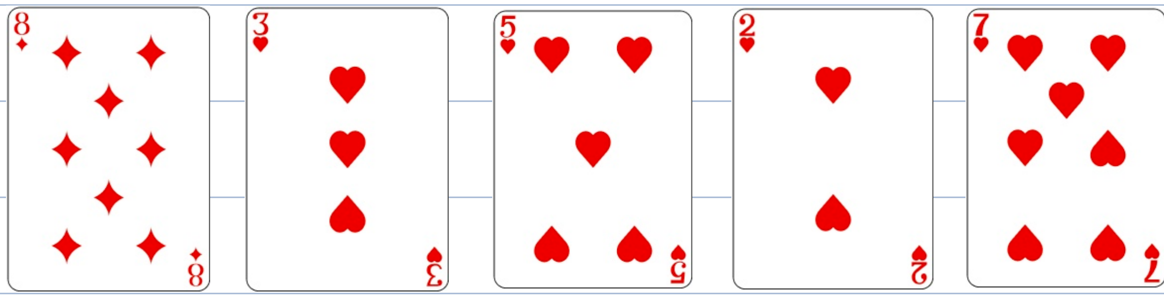
$$\begin{array}{ccccccc} & & 0 & 1 & 2 & 3 & 4 \\ A = [& & \cancel{5} & \cancel{8} & 1 & \cancel{4} & 2\ ] \\ \longrightarrow & & 1 & 2 & \cancel{8} & 5 & 8 \\ & & & & 4 & & \end{array}$$

```
for i → 0 to (N-2) {
    mi = i
    for j → i to (N-1) {
        if (A[j] < A[mi])    mi = j
    }
    swap (A, mi, i)
}
```

TC = $\underline{O(N^2)}$ SC = $\underline{O(1)}$

H.W → Bubble Sort ✓

Insertion Sort (Arrangement of playing cards)



Running stream of integers →

8 10 9 15 5 11

5 ~~8~~	8 ~~9~~	9 ~~10~~	10 ~~15~~	11 ~~15~~	15
0	1	2	3	4	5

```
r = -1    // index of rightmost element in A[]
for i → 0 to (N-1) {    // N → # inputs
    // input → x
    j = r
    while (j >= 0 && A[j] > x) {
            A[j+1] = A[j]
            j--
    }

    A[j+1] = x        r++
}
```

TC = $\underline{O(N^2)}$

SC = $\underline{O(1)}$